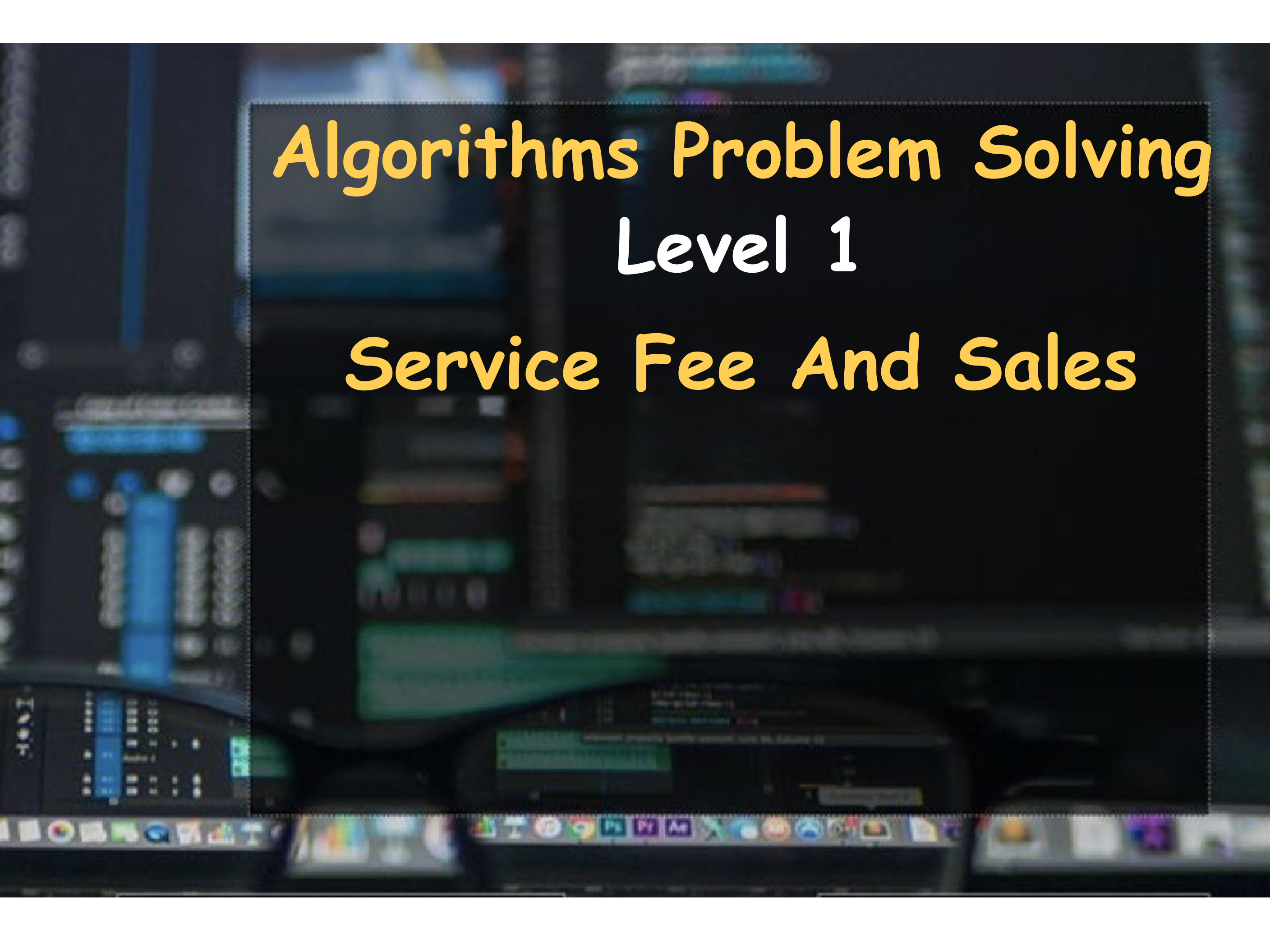




# Algorithms Problem Solving Level 1

## Service Fee And Sales



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## Definition

- ◆ This program reads a positive bill value and adds a 10% service fee followed by a 16% sales tax.
- ◆ The final bill is calculated by applying both percentages to the bill.
- ◆ For an input of 100, the final result is 127.6.

## Explanation

◆ `readPositiveNumb()` keeps asking for the bill value until the user enters a number greater than 0.

◆ `totalBillWithServiceAndTax()` applies the service fee first:

``TotalBill * 1.1``

◆ Multiplying by ``1.1`` increases the value by 10%.

◆ Then it applies the sales tax:

``TotalBill * 1.16``

◆ Multiplying by ``1.16`` increases the value by 16%.

◆ The function returns the final bill.

◆ `printTotalBill()` displays the final amount.

◆ In `main()`, the functions are called from inside each other:

``printTotalBill(totalBillWithServiceAndTax(readPositiveNumb()));``

## Steps in main()

1. Read a positive bill value.
2. Add the 10% service fee.
3. Add the 16% sales tax to the updated amount.
4. Return the final bill.
5. Print the total bill.

# 💡 Example

◆ If the user enters:

Bill Value: 100

◆ After 10% service fee:

$$100 \times 1.10 = 110$$

◆ After 16% sales tax:

$$110 \times 1.16 = 127.6$$

➡ Result

The Total Bill = 127.6

## Note

- ◆ The percentages are applied sequentially.
- ◆ The 16% tax is calculated on the amount after the 10% service fee, not on the original bill.
- ◆ `1.10` represents the original 100% plus 10%.
- ◆ `1.16` represents the original 100% plus 16%.

## ⊘ Warning

◆ The function name `totalBillWithServiceAndTax()` is clear, but the parameter is named `TotalBill` even though it initially represents the original bill value. A name such as `BillValue` would be clearer.



# Comparison

## ▨ Sequential calculation

- ✓ First add 10% service fee.
- ✓ Then add 16% tax to the updated amount.
- ✓ For 100:  $100 \times 1.10 \times 1.16 = 127.6$ .

## ▤ Adding both percentages directly

- ✓  $10\% + 16\% = 26\%$ .
- ✓ Applying 26% directly to 100 gives 126.
- ✓ This is different because the 16% tax in the program is applied after the service fee.

## ✦ Summary

- ✓ The program validates that the bill value is positive.
- ✓ A 10% service fee is applied first.
- ✓ A 16% sales tax is then applied to the updated amount.
- ✓ Sequential percentage calculations can produce a different result from simply adding the percentages.

Thank you 😊